

Differentiation – Product Rule: Worked Examples

Product Rule

If $y = uv$ where u and v are functions of x , then

$$\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

Example

Differentiate:

$$y = 3x^2 e^{4x}$$

$$u = 3x^2; \quad \frac{du}{dx} = 6x$$

$$v = e^{4x}; \quad \frac{dv}{dx} = 4e^{4x}$$

$$\frac{dy}{dx} = 3x^2(4e^{4x}) + e^{4x}(6x)$$

$$\frac{dy}{dx} = 12x^2 e^{4x} + 6x e^{4x}$$

This could be factorised to

$$\frac{dy}{dx} = 6x e^{4x} (2x + 1)$$